Learning-guided candidate selection for compact B-spline fitting

Ordered samples -> dense candidates -> one adaptive mask -> survivor update -> one final refit

1 Geometry + parameters

Ordered points $Q \in \mathbb{R}^{N \times 2}$



GeometryEncoder

Coordinates + chord derivatives
3 convolution blocks; width 128

Point features F + pooled feature g

ParameterHead

Chord reference + learned residual
Curve-level learned trust

$0 = t_0 < t_1 < \cdots < t_{N-1} = 1$

2 Candidate proposal

CandidateKnotHead

Local Gaussian cross-attention
K/V: point features + PosEnc(t)

$K_c + 1$ interval queries $\rightarrow K_c$ knots



Ordered candidates + node tokens

Base proposal decode

One t / U update; no extra stack
Parameter warp + bounded knot shift

$K_c = 32$; point-node interaction

3 Adaptive selection

Interactive selector

Attention blocks x 2
No extra selection-refinement stack

$p_j = \sigma(s_j - \bar{s} - \beta)$

Curve-conditioned threshold β

Probability-mass Top-K

Minimum 4 knots; 4 coverage bins
Rank once; build one discrete mask



KeepMask $m \in \{0,1\}^{K_c}$

4 Survivor refinement

Selected-node decoder

K/V: retained nodes only
Neighbor gaps + rank + selected count

Base survivor-conditioned decoder

Parameter / knot update once

Refresh positions and point features
Monotone parameter update
Bounded survivor relocation



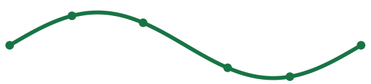
Updated t^*, U^*; same mask m

5 B-spline output

Standard cubic B-spline

$P^* = \arg\min_P \|B(t^*, U^*[m])P - Q\|_F^2$

$P_0 = Q_0,\ P_n = Q_{N-1}$; refit ×1



Fitted curve + control vertices

Internal knots + measured errors

Open-clamped output
one forward; no greedy search

TRAINING ONLY

Synthetic labels

True parameters + knot positions

Counterfactual / compact-subset teacher

Try masks / geometry; evaluate actual B-spline fits

Joint supervision

Fit + keep / count / geometry; MaxSE checked at deployment

Fixed-depth learned refinement; discrete selection is made once. Feasibility is measured after the final fit.